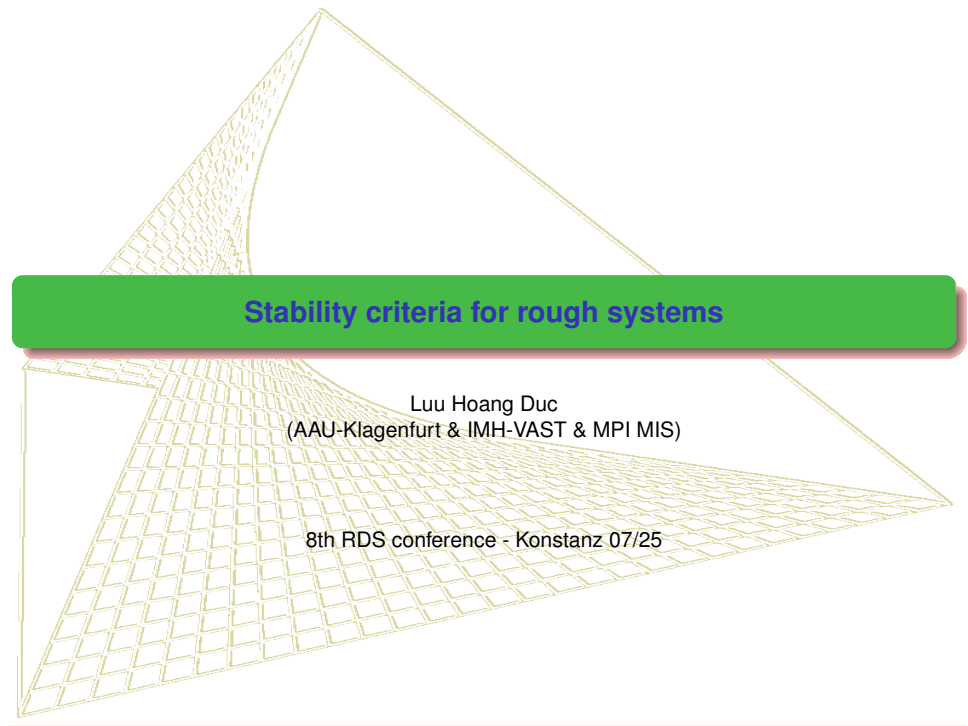




Stability criteria for rough systems

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joint work with Nguyen Dinh Cong, Phan Thanh Hong, Jürgen Jost

Stochastic stability

- Asymptotic stability for deterministic systems: Lyapunov (1892) $\epsilon - \delta$ language.
- Stochastic stability for Markov systems: Khasminskii (1979), Skorohod (1987), Molchanov, Arnold, Pardoux, Baxendale, Mao... and many others in control theory.

$$dy_t = f(y_t)dt + g(y_t)dB_t, \quad (1.1)$$

$f : \mathbb{R}^d \rightarrow \mathbb{R}^d, g : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$ glob. Lischitz continuous w.r.t. C_f, C_g and f strictly dissipative, i.e. $\langle y, f(y) \rangle \leq -d\|y\|^2$. By Itô's formula

$$d\mathbb{E}\|y_t\|^2 = \mathbb{E}\left(2\langle y_t, f(y_t) \rangle + \|g(y_t)\|^2\right)dt + \mathbb{E}2\langle y_t, g(y_t)dB_t \rangle \quad (1.2)$$

$$\leq -(2d - C_g^2)\mathbb{E}\|y_t\|^2 dt. \Rightarrow C_g^2 < 2d! \quad (1.3)$$

- Later generalize to Lyapunov function $V : \mathbb{R}^d \rightarrow \mathbb{R}_+$ and $\exists d_1, d_2 > 0$

$$\langle \nabla V(z), f(z) \rangle \leq d_1 - d_2 V(z), \quad \forall z \in \mathbb{R}^d.$$

- When B is replaced by B^H for $H \in (\frac{1}{3}, \frac{1}{2})$, one needs Lyons' rough path theory and more regularity, e.g. $(H_g) g$ is in $C_b^3(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^m, \mathbb{R}^d))$ with $C_g = \max\{\|D^{(i)}g\|_\infty : i = 0, \dots, 3\}$.
- Ergodicity theory by Mattingly, Hairer, Scheutzow, Meyn, Tweedie,... even for B^H .
- Related topics on RDS and random (numerical) attractors: Crauel, Kloeden, Arnold, Duan, Schmalfuss, Garrido-Atienza,... Even more for SPDEs with the semi-group technique.

Doss-Sussmann transformation

- Rough ODE $dy_t = f(y_t)dt + g(y_t)d\mathbf{x}_t$ is understood in the Gubinelli rough integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)d\mathbf{x}_s. \quad (1.4)$$

- the solution $\phi(\mathbf{x}, \phi_a)$ of $d\phi_u = g(\phi_u)d\mathbf{x}_u$, $u \in [a, b]$, $\phi_a \in \mathbb{R}^d$ is C^1 w.r.t. ϕ_a , and $\frac{\partial \phi}{\partial \phi_a}(\cdot, \mathbf{x}, \phi_a)$ is the solution of the linearized system

$$d\xi_u = Dg(\phi_u(\mathbf{x}, \phi_a))\xi_u d\mathbf{x}_u, \quad u \in [a, b], \xi_a = Id, \quad (1.5)$$

- Cass-Litterer-Lyons (2013): Stopping times

$$\tau_0 = a, \quad \tau_{i+1} := \inf \left\{ t > \tau_i : \|\mathbf{x}\|_{p\text{-var}, [\tau_i, t]} = \gamma \right\} \wedge b.$$

- The Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t) = z_t + \int_{\tau_i}^t g(\phi_s(\mathbf{x}, z_t))d\mathbf{x}_s$ on $[\tau_i, \tau_{i+1}]$

$$\dot{z}_t = \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} f(\phi_t(\mathbf{x}, z_t)) = (I + \psi_t)f(z_t + \eta_t), \quad t \in [\tau_i, \tau_{i+1}], \quad z_{\tau_i} = y_{\tau_i}. \quad (1.6)$$

$$\eta_t := y_t - z_t, \quad \text{and} \quad \psi_t := \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} - Id, \quad \forall t \in [\tau_i, \tau_{i+1}].$$

Random attractors for RDE: finite dimension

- Assumptions for f : $(\mathbf{H}'_f) : \exists d_1 \geq 0, d_2 > 0 : \langle y, f(y) \rangle \leq d_1 - d_2 \|y\|^2, \quad \forall y \in \mathbb{R}^d$; and

$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad \forall y \neq 0; \quad (1.7)$$

E.g. f Lipschitz or $f(y) = y[\alpha - h(\|y\|)]$.

- Key:** $\gamma := \frac{\lambda}{16C_p C_g}$ for $\lambda < 1$ then $\|\eta\|_{\infty, [\tau_i, \tau_{i+1}]}, \|\psi\|_{\infty, [\tau_i, \tau_{i+1}]} \leq \lambda$
- Compactness arguments: from $\dot{z}_t = (Id + \psi_t)f(z_t + \eta_t)$, $\exists \bar{C}_\lambda, \delta_\lambda > 0$ such that

$$\frac{d}{2dt} \|z_t\|^2 \leq \bar{C}_\lambda - \delta_\lambda \|z_t\|^2, \quad \forall t \in [\tau_i, \tau_{i+1}]. \quad (1.8)$$

Hence $\|y_t\| \leq \|y_0\| e^{-\delta_\lambda t} + C_\lambda N\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [0, t]\right)$ for $N(\gamma, \mathbf{x}, [a, b]) := \sup\{i : \tau_i \leq b\}$.

Theorem (LHD-DCDS 2022)

Under $(\mathbf{H}'_f), (\mathbf{H}_g), (\mathbf{H}_X)$, there exists a random pullback attractor $\mathcal{A}(\omega)$ such that $|\mathcal{A}(\cdot)| \in \mathcal{L}^p$ for any $\rho \geq 1$. If f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity. Moreover, if f is strictly dissipative then $\mathcal{A}(\omega)$ is a singleton, provided that C_g is sufficiently small.

Numerical attractors for discrete systems

$$y_0^\Delta \in \mathbb{R}^d,$$

$$y_{t_{k+1}}^\Delta = y_{t_k}^\Delta + f(y_{t_k}^\Delta)\Delta + g(y_{t_k}^\Delta)x_{t_k, t_{k+1}} + Dg(y_{t_k}^\Delta)g(y_{t_k}^\Delta)\mathbb{X}_{t_k, t_{k+1}}, \quad k \in \mathbb{N}. \quad (1.9)$$

under the regular grid $\Pi = \{t_k := k\Delta\}_{k \in \mathbb{N}}$, $0 < \Delta \leq 1$.

Theorem (LHD, P. Kloeden (SINUM 2023); LHD, N.D. Cong, P. T. Hong (JDDE 2023))

Assume that $y(t, 0, y_0)$ is a solution of the rough differential equation (1.4) on $[0, T]$. Then there exists a generic constant $C = C(f, g, \|\mathbf{x}\|_{\nu, [0, T]}, T, \|y_0\|) > 0$ such that for $\Delta \ll 1$

$$\sup_{0 \leq k \leq m} \|y(t_k, 0, y_0) - y_k^\Delta\| \leq C\Delta^{3\nu-1}. \quad (1.10)$$

Assume further that f is globally Lipschitz continuous and dissipative. Then φ^h admits a numerical pullback attractor $\mathcal{A}^h \in \mathcal{L}^\rho$ which converges to the attractor $\mathcal{A}$ a.s.

$$\lim_{h \rightarrow 0} d_H(\mathcal{A}^h | \mathcal{A}) = 0 \quad \text{a.s.} \quad (1.11)$$

Compactness arguments:

$$\|y_1\|^2 \leq (\|\bar{y}_1(y_0)\| + \xi_1(\mathbf{x}, \Delta)\|y_0\|^\beta + \xi_2(\mathbf{x}, \Delta))^2 \leq \left[(1 + \varepsilon)e^{-d} + \varepsilon\right]\|y_0\|^2 + \bar{\xi}(\mathbf{x}, \Delta).$$

Fitzhugh-Nagumo neuro-system

The deterministic FHN neuron model is

$$\begin{cases} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon(v_t - \mu w_t + J) \end{cases} \quad (1.12)$$

f is dissipative but does not satisfy

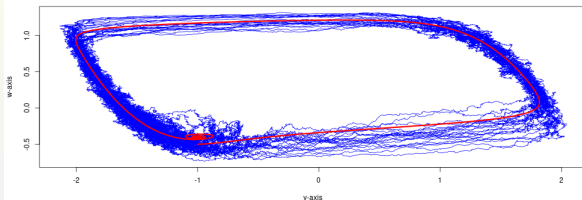
$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad (1.13)$$

Instead, we introduce a Lyapunov function

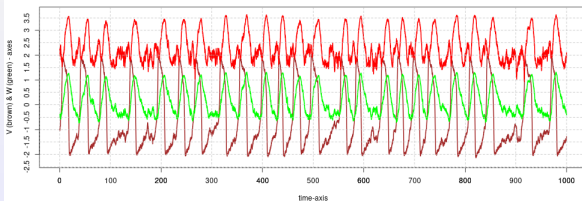
$$V(y) := \left(1 + v^4 + Bw^2\right)^{\frac{1}{4}}, \quad (1.14)$$

$$\|\nabla V\| \leq L_V. \quad (1.15)$$

Trajectory of rough FHN system. Sigma = 0.09



Time series rough FHN system



Lyapunov function

Classical condition on the gradient ∇V along the vector field f : $\exists d_1, d_2 > 0$

$$\langle \nabla V(z), f(z) \rangle \leq d_1 - d_2 V(z); \Rightarrow \langle \nabla V(z), (I + \psi)f(z + \eta) \rangle \leq d_1 - d_2 V(z) \quad \forall [\tau_i, \tau_{i+1}]? \quad (1.16)$$

But η_t, ψ_t varies as stochastically as x_t , e.g. $\eta_t = \int_{\tau_i}^t g(\phi_s(\mathbf{x}, z_t)) dB_s(\mathbf{x})$.

($\mathbf{H}_V$) There exists for a Lyapunov-type function V for the drift f that satisfies

- there exists functions $\alpha, \beta \in \mathcal{K}'_\infty$ such that

$$\alpha(\|z\|) \leq V(z) \leq \beta(\|z\|), \quad \forall z \in \mathbb{R}^d; \quad (1.17)$$

- there exists a constant $L_V > 0$ such that

$$\|\nabla V(z)\| \leq L_V, \quad \forall z \in \mathbb{R}^d; \quad (1.18)$$

- there exists a sufficiently small constant $\lambda \in (0, 1)$ and constants $C_\lambda, \delta > 0$ such that

$$\sup_{\|\psi\|, \|\eta\| \leq \lambda} \langle \nabla V(z), (I + \psi)f(z + \eta) \rangle \leq C_\lambda - \delta V(z), \quad \forall z \in \mathbb{R}^d. \quad (1.19)$$

① (1.19) is satisfied for a range $0 \leq \lambda < \lambda_0$ with C_{λ_0}, δ . **No g involved !**

② If f is globally Lipschitz, then (1.16) yields (1.19).

③ Examples: $V(z) := \sqrt{1 + \|z\|^2}$ for dissipative f ; or $V(z) := \sqrt{1 + \frac{1}{2}w^2 + \sigma^2(1 - \cos v)}$ for pendulum; or FHN system $V(z) = \left(1 + v^4 + Bw^2\right)^{\frac{1}{4}}$.

Lyapunov functions

Compact arguments:

$$V(y_t) - \frac{C_\lambda}{\delta} \leq e^{-\delta t} \left[V(y_0) - \frac{C_\lambda}{\delta} \right] + H(\|\mathbf{x}\|_{p\text{-var},[0,t]}), \quad \forall t \in [0, T] \quad (1.20)$$

$$H(\xi) = L_V(16C_p C_g)^p \lambda^{1-p} \xi^p + 8L_V C_p C_g \xi. \quad (1.21)$$

$$\bar{R}_\lambda(\omega) := \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\theta_{-k}\omega)\|_{p\text{-var},[-1,0]}) = \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\omega)\|_{p\text{-var},[-1-k,-k]}). \quad (1.22)$$

Theorem (Random pullback attractor)

$$\mathcal{A}(\omega) = \bigcap_{n=1}^{\infty} \varphi(n, \theta_{-n}\omega) \bar{B}^\varepsilon(\theta_{-n}\omega) = \lim_{n \rightarrow \infty} \varphi(n, \theta_{-n}\omega) \bar{B}^\varepsilon(\theta_{-n}\omega), \quad \forall \omega \in \Omega. \quad (1.23)$$

$$\bar{B}^\varepsilon(\omega) := V^{-1}\left(\frac{C_\lambda}{\delta} + \bar{R}_\lambda(\omega) + \varepsilon\right), \quad \forall \varepsilon \geq 0. \quad (1.24)$$

Motivations

- Stability criterion should be free of diffusion g , at most only C_g is needed.
- Data-driven approach: only f (with parameters) and realizations α are given.
- (LHD-PT Hong-ND Cong: SICON 2019): Linearization

$$d\xi = Df(\alpha_t)\xi dt + Dg(\alpha_t)\xi dx = A(t)\xi dt + C(t)\xi dx.$$

$$\xi_t = S(t, \tau, \alpha)\xi_0 + \int_{\tau}^t S(t, s, \alpha) Dg(\alpha_s)\xi_s dx_s. \quad (2.1)$$

Spectral theory works only if g is known!

- Discrete system, even Young equations: $\xi_{k,k+1} = Df(\alpha_k)\xi_k \Delta + Dg(\alpha_k)\xi_k \Delta x_k$?

Settings

Define $\ell(f, \hat{y}, r) := \sup_{0 < \|y - \hat{y}\| < r} \left\langle \frac{y - \hat{y}}{\|y - \hat{y}\|}, \frac{f(y) - f(\hat{y})}{\|y - \hat{y}\|} \right\rangle$ and $\ell(f, \hat{y}) := \lim_{r \rightarrow 0} \ell(f, \hat{y}, r)$. (2.2)

Then $\ell(f, \hat{y}) = \sup_{\|h\|=1} \langle h, Df(\hat{y})h \rangle, \quad \forall \hat{y} \in \mathbb{R}^d;$

$\langle y - \hat{y}, f(y) - f(\hat{y}) \rangle \leq M(Df, \hat{y}, r) \|y - \hat{y}\|^2, \quad \forall \|y - \hat{y}\| < r.$ (2.3)

Introduce the function

$$\begin{aligned} \kappa(\lambda, y, r) &:= M(Df, y, r)(1 + 16\lambda^2) + 8\lambda|M(Df, y, r)| + K(\lambda)\|f(y)\| \\ &\quad + 8\lambda(1 + 4\lambda)\|Df\|_{\infty, B(y, r)}, \\ \kappa(\lambda, y, 0) &= \ell(f, y)(1 + 16\lambda^2) + 8\lambda|\ell(f, y)| + K(\lambda)\|f(y)\| + 8\lambda(1 + 4\lambda)\|Df(y)\| \end{aligned} \quad (2.4)$$

Observation: $\kappa(\lambda, y, 0) \approx \ell(f, y)$ for $\lambda \ll 1$.

$$C_g := \max \left\{ \|Dg\|_{\infty}, \sqrt{\|Dg\|_{\infty} \|g\|_{\infty}}, \sqrt{\|D^2g\|_{\infty} \|g\|_{\infty}}, \sqrt{\|D^3g\|_{\infty} \|g\|_{\infty}}, \right. \\ \left. \left(\|D^2g\|_{\infty} \|g\|_{\infty}^2 \right)^{\frac{1}{3}}, \left(\|D^3g\|_{\infty} \|g\|_{\infty}^2 \right)^{\frac{1}{3}} \right\}. \quad (2.5)$$

$$L_g := \max \left\{ \|Dg\|_{\infty}, \|D^2g\|_{\infty}, \left[\|g\|_{\infty} (\|Dg\|_{\infty} \vee \|D^2g\|_{\infty} \vee \|D^3g\|_{\infty}) \right]^{\frac{1}{2}}, \left(\|g\|_{\infty}^2 \|D^3g\|_{\infty} \right)^{\frac{1}{3}} \right\}.$$

Main results

Theorem (Continuous time)

Let $\alpha(\cdot)$ be a stationary solution that is L^p -integrable. If the linearization $\dot{\xi} = Df(\alpha_t)\xi$ is exponentially stable, then α_t is exponentially stable a.s. as long as C_g is sufficiently small and satisfies a stability criterion: $\exists \lambda \in (0, \frac{1}{8})$ such that

$$-\mathbb{E}\kappa(\lambda, \alpha(\cdot), 0) > 4\lambda \mathbb{E}N^*\left(\frac{\lambda}{C_p C_g}, \mathbf{x}(\cdot), [0, 1]\right). \quad (2.7)$$

Then $\alpha(\theta_t \omega)$ is exponentially stable a.s.

Simplified version: $-\mathbb{E}\ell(f, \alpha(\cdot), 0) > 4C_g \mathbb{E}N^*\left(\frac{1}{C_p}, \mathbf{x}(\cdot), [0, 1]\right)$. LHS f and α , RHS C_g and $\mathbf{x}$!

Theorem (Discrete time)

Assume $L_f := \|Df\|_\infty < \infty$, (H_g) , (H_X) , $\|\alpha^\Delta(\cdot)\| \in L^p$ and there exist $\lambda \in (0, \frac{1}{2})$, $\gamma^* \in (0, 1)$

$$\eta := -\mathbb{E}\ell(f, \alpha^\Delta(\cdot)) - \frac{1}{2}L_f^2\Delta > KL_g\gamma^* \left[\mathbb{E}N^*(\gamma^*, \mathbf{x}(\cdot), [0, 1]) + 2\mathbb{E}N^*(\lambda, S, [0, 1]) \right]. \quad (2.8)$$

Then $\alpha^\Delta(\cdot)$ is exponentially stable a.s.

Sketch of the proof

Lemma (LHDuc (PAFA & DCDS -2022))

$\lambda := C_p C_g \|\mathbf{x}\|_{p\text{-var}, [a, b]} \leq 1/8$. Then

$$\|\eta\|_{\infty, [a, b]} \leq 2(\|g\|_{\infty} \|\mathbf{x}\|_{p\text{-var}, [a, b]} + \lambda^2); \quad \|\psi\|_{\infty, [a, b]} \leq 4\lambda \quad (2.9)$$

Consider $y_t = \phi_t(\mathbf{x}, z_t)$ and $\bar{y}_t = \phi_t(\mathbf{x}, \bar{z}_t)$. Assign

$$\eta_t = y_t - z_t; \quad \bar{\eta}_t = \bar{y}_t - \bar{z}_t; \quad \psi_t = \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} - Id; \quad \bar{\psi}_t = \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, \bar{z}_t) \right]^{-1} - Id;$$

$\exists K : [0, 1] \rightarrow \mathbb{R}_+$ with $K(0) = 0$, such that

$$\|\bar{\eta}_t - \eta_t\| \leq 4\lambda \|\bar{z}_t - z_t\|; \quad \|\bar{\psi}_t - \psi_t\| \leq K(\lambda) \|\bar{z}_t - z_t\|, \quad \forall t \in [a, b]. \quad (2.10)$$

Sketch of the proof

Let z_t be a the solution of the associate ODE (1.6) starting at y_0

$$\dot{z}_t = \left[\frac{\partial \phi}{\partial Z}(t, \mathbf{x}(\omega), z_t) \right]^{-1} f(\phi_t(\mathbf{x}(\omega), z_t)), \quad t \in [0, \tau], \quad z_0 = y_0,$$

and similarly let $\bar{z}_t$ be the solution of this ODE starting at $\bar{z}_0 = \alpha(\omega)$ where $\alpha_t = \phi(t, \mathbf{x}(\omega), \bar{z}_t) = \bar{z}_t + \bar{\eta}_t$. Consider

$$\eta_t = y_t - z_t; \quad \bar{\eta}_t = \alpha_t - \bar{z}_t; \quad \psi_t = \left[\frac{\partial \phi}{\partial Z}(t, \mathbf{x}(\omega), z_t) \right]^{-1} - Id; \quad \bar{\psi}_t = \left[\frac{\partial \phi}{\partial Z}(t, \mathbf{x}(\omega), \bar{z}_t) \right]^{-1} - Id.$$

and the difference $\gamma_t := z_t - \bar{z}_t$, then

$$\begin{aligned} \|\bar{\eta}_t - \eta_t\| &\leq 4\lambda \|\bar{z}_t - z_t\| \leq 4\lambda \|\gamma_t\|; \\ \|\bar{\psi}_t - \psi_t\| &\leq K(\lambda) \|\bar{z}_t - z_t\| \leq K(\lambda) \|\gamma_t\|; \\ (1 - 4\lambda) \|\gamma_t\| &\leq \|y_t - \alpha_t\| \leq (1 + 4\lambda) \|\gamma_t\|; \quad \forall t \in [0, \tau]. \end{aligned}$$

Hence γ evolves with its equation

$$\begin{aligned} \dot{\gamma}_t &= \dot{z}_t - \dot{\bar{z}}_t \\ &= (Id + \psi_t)f(y_t) - (Id + \bar{\psi}_t)f(\alpha_t) \\ &= [f(y_t) - f(\alpha_t)] + \psi_t[f(y_t) - f(\alpha_t)] + (\psi_t - \bar{\psi}_t)f(\alpha_t). \end{aligned}$$

Sketch of the proof

It follows from Lagrange's mean value theorem that for all $t \in [0, \tau]$ and $\|y_t - a_t\| < r$,

$$\begin{aligned} \frac{d}{dt} \|\gamma_t\|^2 &\leq 2 \langle y_t - a_t + (\bar{\eta}_t - \eta_t), f(y_t) - f(a_t) \rangle + \dots \\ &\leq 2 \langle y_t - a_t, f(y_t) - f(a_t) \rangle + \dots \\ &\leq 2M(Df, a_t, r) \|y_t - a_t\|^2 + 2\|\gamma_t\| \|\psi_t - \bar{\psi}_t\| \|f(a_t)\| \\ &\quad + 2 \left(\|\eta_t - \bar{\eta}_t\| + \|\gamma_t\| \|\psi_t\| \right) \|Df\|_{\infty, B(a_t, r)} \|y_t - a_t\|. \end{aligned}$$

As a result, $\frac{d}{dt} \|\gamma_t\|^2 \leq 2\kappa(\lambda, a_t, r) \|\gamma_t\|^2$, which yields

$$\|y_t - a_t\| \leq (1 + 4\lambda) \|\gamma_t\| \leq \exp \left\{ 4\lambda + \int_0^t \kappa(\lambda, a_s, r) ds \right\} \|y_0 - a(\omega)\|, \quad \forall t \in [0, \tau]. \quad (2.11)$$

By induction for stopping times and a local arguments, we can choose

$$\|y_0 - a(\omega)\| \leq R(\omega) \quad := \quad r \inf_{n \geq 0} \left[\inf_{t \in [\tau_n, \tau_{n+1}]} \exp \left\{ -4\lambda(n+1) - \int_0^t \kappa(\lambda, a_s, r) ds \right\} \right]. \quad (2.12)$$

Assumptions and settings

Throughout this section, we will assume $(\mathbf{H}_f)$, $(\mathbf{H}_g)$, $(\mathbf{H}_X)$ and further more:

$$L_f := \|Df\|_\infty < \infty. \quad (2.13)$$

Recall that

$$\langle y - a^\Delta(\omega), f(y) - f(a^\Delta(\omega)) \rangle \leq M(Df, a^\Delta(\omega), r) \|y - a^\Delta(\omega)\|^2, \quad \forall \|y - a^\Delta(\omega)\| < r, \forall \omega \in \Omega \quad (2.14)$$

Assign $h_t := y_t^\Delta - a_t^\Delta$ and

$$P_{s,t} := [g(y_s^\Delta)x_{s,t} + Dg(y_s^\Delta)g(y_s^\Delta)\mathbb{X}_{s,t}] - [g(a_s^\Delta)x_{s,t} + Dg(a_s^\Delta)g(a_s^\Delta)\mathbb{X}_{s,t}].$$

Then

$$h_{t_{k+1}} = h_{t_k} + [f(y_{t_k}^\Delta) - f(a_{t_k}^\Delta)]\Delta + P_{t_k, t_{k+1}}. \quad (2.15)$$

and it is easy to check that

$$\|P_{s,t}\| \leq \left[\|Dg\|_\infty \|x_{s,t}\| + (\|g\|_\infty \|D^2g\|_\infty + \|Dg\|_\infty^2) \|\mathbb{X}_{s,t}\| \right] \cdot \|h_s\|. \quad (2.16)$$

Observations

Observe that whenever $\|y_{t_k}^\Delta - a_{t_k}^\Delta\| < r$,

$$\begin{aligned}
 \|h_{t_{k+1}}\|^2 &= \|h_{t_k}\|^2 + \|f(y_{t_k}^\Delta) - f(a_{t_k}^\Delta)\|^2 \Delta^2 + \|P_{t_k, t_{k+1}}\|^2 \\
 &\quad + 2\langle y_{t_k}^\Delta - a_{t_k}^\Delta, f(y_{t_k}^\Delta) - f(a_{t_k}^\Delta) \rangle \Delta + 2\langle h_{t_k}, P_{t_k, t_{k+1}} \rangle + 2\langle f(y_{t_k}^\Delta) - f(a_{t_k}^\Delta), P_{t_k, t_{k+1}} \rangle \Delta \\
 &\leq \|h_{t_k}\|^2 \left[1 + 2M(a_{t_k}^\Delta) \Delta + L_f^2 \Delta^2 \right] + 2\langle f(y_{t_k}^\Delta) - f(a_{t_k}^\Delta), P_{t_k, t_{k+1}} \rangle \Delta \\
 &\quad + 2\langle h_{t_k}, P_{t_k, t_{k+1}} \rangle + \|P_{t_k, t_{k+1}}\|^2.
 \end{aligned} \tag{2.17}$$

Consider the first interval $[\tau_0^\Delta, \tau_1^\Delta]$ where $\tau_0^\Delta = 0$. There are two possibilities. The trivial case is when

$$\left[L_f(\tau_1^\Delta - \tau_0^\Delta) + 2\lambda \|f(a^\Delta)\|_{\infty, \Pi[\tau_0^\Delta, \tau_1^\Delta]}(\tau_1^\Delta - \tau_0^\Delta) + (4C_p + 1)L_g \|\mathbf{x}\|_{p\text{-var}, \Pi[\tau_0^\Delta, \tau_1^\Delta]} \right] > \lambda,$$

then $\tau_0^\Delta, \tau_1^\Delta$ are consecutive in Π , i.e. $\tau_1^\Delta - \tau_0^\Delta = \Delta$. Then (2.16) and (2.17) yield

$$\|h_{\tau_1^\Delta}\|^2 \leq \exp \left\{ \left[2M(a_{\tau_0^\Delta}^\Delta) + L_f^2 \Delta \right] (\tau_1^\Delta - \tau_0^\Delta) + 3e^{6L_f \Delta} L_g \|\mathbf{x}\|_{p\text{-var}, [\tau_0^\Delta, \tau_1^\Delta]} \right\} \|h_{\tau_0^\Delta}\|^2$$

as $\|h_{\tau_0^\Delta}\| \leq r$ by assumption.

Auxiliary results

Lemma

If $4C_p L_g \|\mathbf{x}\|_{p\text{-var}, \Pi[a,b]} \leq \lambda < \frac{1}{2}$, the following estimate holds

$$\begin{aligned} \|h, R^h\|_{p\text{-var}, \Pi[a,b]} &\leq \left[L_f(b-a) + 2\lambda \|f(\mathfrak{a}^\Delta)\|_{\infty, [a,b]}(b-a) + (4C_p + 1)L_g \|\mathbf{x}\|_{p\text{-var}, \Pi[a,b]} \right] \\ &\quad \times \|h, R^h\|_{p\text{-var}, \Pi[a,b]}. \end{aligned}$$

This motivates us to construct a discrete sequence of stopping times $\tau_n^\Delta(\lambda, \mathcal{S}, [0, \infty))$

$$\mathcal{S} = \{\mathbf{w}^{(1)}, \mathbf{w}^{(2)}, \mathbf{w}^{(3)}, \mathbf{w}^{(4)}\},$$

in which

$$\begin{aligned} \mathbf{w}_{s,t}^{(1)} &= L_f(t-s), \beta_1 = 1; & \mathbf{w}_{s,t}^{(2)} &= (4C_p + 1)^p L_g^p \|x\|_{p\text{-var}, \Pi[s,t]}^p, \beta_2 = \frac{1}{p}; \\ \mathbf{w}_{s,t}^{(3)} &= (4C_p + 1)^p L_g^p \|\mathbb{X}\|_{q\text{-var}, \Pi[s,t]}^q, \beta_3 = \frac{1}{q}; & \mathbf{w}_{s,t}^{(4)} &= 2\lambda \|f(\mathfrak{a}^\Delta)\|_{\infty, \Pi[s,t]}(t-s), \beta_4 = 1. \end{aligned}$$

Stopping time analysis

- **Continuous time:** for any fixed $\gamma \in (0, 1)$ the sequence of stopping times $\{\tau_i(\gamma, \mathbf{x}, [0, \infty))\}_{i \in \mathbb{N}}$ is defined by

$$\tau_0 = 0, \quad \tau_{i+1} := \inf \left\{ t > \tau_i : \|\mathbf{x}\|_{p\text{-var}, [\tau_i, t]} = \gamma \right\}. \quad (2.18)$$

For a fixed closed interval $I \subset [0, \infty)$, we define another sequence of stopping times $\{\tau_i^*(\gamma, \mathbf{x}, I)\}_{i \in \mathbb{N}}$ by

$$\tau_0^* = \min I, \quad \tau_{i+1}^* := \inf \left\{ t > \tau_i^* : \|\mathbf{x}\|_{p\text{-var}, [\tau_i^*, t]} = \gamma \right\} \wedge \max I. \quad (2.19)$$

Define $N^*(\gamma, \mathbf{x}, I) := \sup\{i \in \mathbb{N} : \tau_i^* < \max I\} + 1$.

- **Discrete time:** For given $\Delta > 0$, consider the sequence $t_k = k\Delta$, $k \in \mathbb{N}$. Given a fixed $\gamma > 0$, assign the starting time $\tau_0^\Delta(\gamma, \mathcal{S}, [0, \infty)) = 0$. For each $n \in \mathbb{N}$, assume $\tau_n^\Delta(\gamma, \mathcal{S}, [0, \infty)) = t_k$ is determined. Then one can define $\tau_{n+1}^\Delta(\gamma, \mathcal{S})$ by the following rule:
 - if $\sum_{\mathbf{w} \in \mathcal{S}} \mathbf{w}_{t_k, t_{k+1}}^{\beta \mathbf{w}} > \gamma$ then set $\tau_{n+1}^\Delta(\gamma, \mathcal{S}, [0, \infty)) := t_{k+1}$;
 - else set $\tau_{n+1}^\Delta(\gamma, \mathcal{S}, [0, \infty)) := \sup\{t_l > t_k : \sum_{\mathbf{w} \in \mathcal{S}} \mathbf{w}_{t_k, t_l}^{\beta \mathbf{w}} \leq \gamma\}$

Observations

Lemma

Assume $\|h_{\tau_0^\Delta}\| \leq \frac{r}{16(1+C_p)}$. If

$$\left[L_f(\tau_1^\Delta - \tau_0^\Delta) + 2\lambda \|f(a^\Delta)\|_{\infty, \Pi[\tau_0^\Delta, \tau_1^\Delta]}(\tau_1^\Delta - \tau_0^\Delta) + (4C_p + 1)L_g \|\mathbf{x}\|_{p-\text{var}, \Pi[\tau_0^\Delta, \tau_1^\Delta]} \right] \leq \lambda$$

then by induction in t_k and discrete sewing lemma

$$\|h_{\tau_1^\Delta}\|^2 \leq \exp \left\{ \left[2\bar{M}_0 + L_f^2 \Delta \right] (\tau_1^\Delta - \tau_0^\Delta) + 6(1+\lambda)(1+2C_p)^2 e^{12\lambda+2\lambda^2} L_g \|\mathbf{x}\|_{p-\text{var}, \Pi[\tau_0^\Delta, \tau_1^\Delta]} \right\} \|h_{\tau_0^\Delta}\|^2. \quad (2.20)$$

where $\bar{M}_0$ denotes the average of $M(a_{t_k}^\Delta)$, $t_k \in [\tau_0^\Delta, \tau_1^\Delta)$.

In general by induction, if $\|h_{\tau_n^\Delta}\| \leq \frac{r}{16(1+C_p)}$ then

$$\|h_{\tau_{n+1}^\Delta}\| \leq \exp \left\{ \left[\bar{M}_n + \frac{1}{2} L_f^2 \Delta \right] (\tau_{n+1}^\Delta - \tau_n^\Delta) + KL_g \gamma^* N^*(\gamma^*, \mathbf{x}, [\tau_n^\Delta, \tau_{n+1}^\Delta]) \right\} \|h_{\tau_n^\Delta}\| \quad (2.21)$$

where $\bar{M}_n$ denotes the average of $M(a_{t_k}^\Delta)$, $t_k \in [\tau_n^\Delta, \tau_{n+1}^\Delta)$ and $\gamma^* \in (0, 1)$.

Then a local arguments ...

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Thank you!